



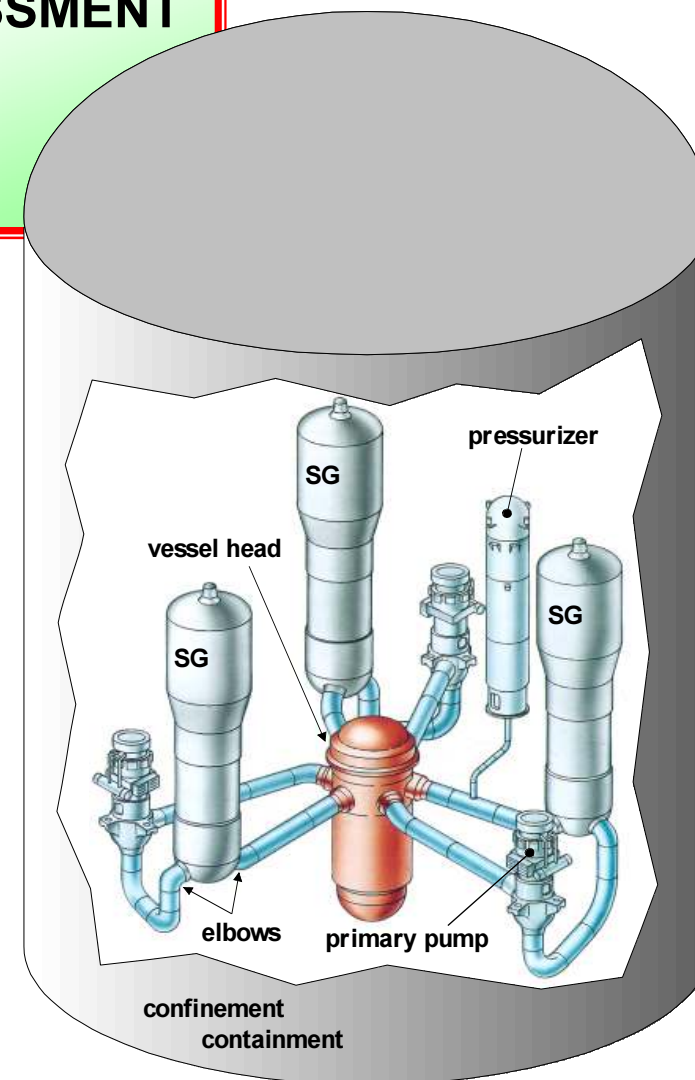
FRENCH NUCLEAR PLANT LIFE MANAGEMENT STRATEGY

REACTOR PRESSURE VESSEL ASSESSMENT AND STEAM GENERATORS LIFE MANAGEMENT STRATEGY

CONTENT

- ⇒ REACTOR PRESSURE VESSEL PLIM
- ⇒ STEAM GENERATORS STRATEGY

Georges BEZDIKIAN





NUCLEAR PLANTS LIFE MANAGEMENT INTRODUCTION

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The capability of nuclear electricity generation by French Utility's (EDF Electricité de France) is 80% of global generation, 58 PWR units are in operation, 34 units three-loop, 20 units four-loop (1300MWe) and 4 units four-loop (1450MWe).

- **The Status of Nuclear Plants Life Management is depending of the following objectives :**
 - **To maintain current operating performances (safety, availability, costs, security, environment) in the long term, and possibly improve on some aspects ;**
 - **wherever possible, to operate the units throughout their design lifetime, in other words 40 years, and after 40 years ;**
 - **To consolidate Mormal and Exceptionnal Maintenance and long term Prediction.**

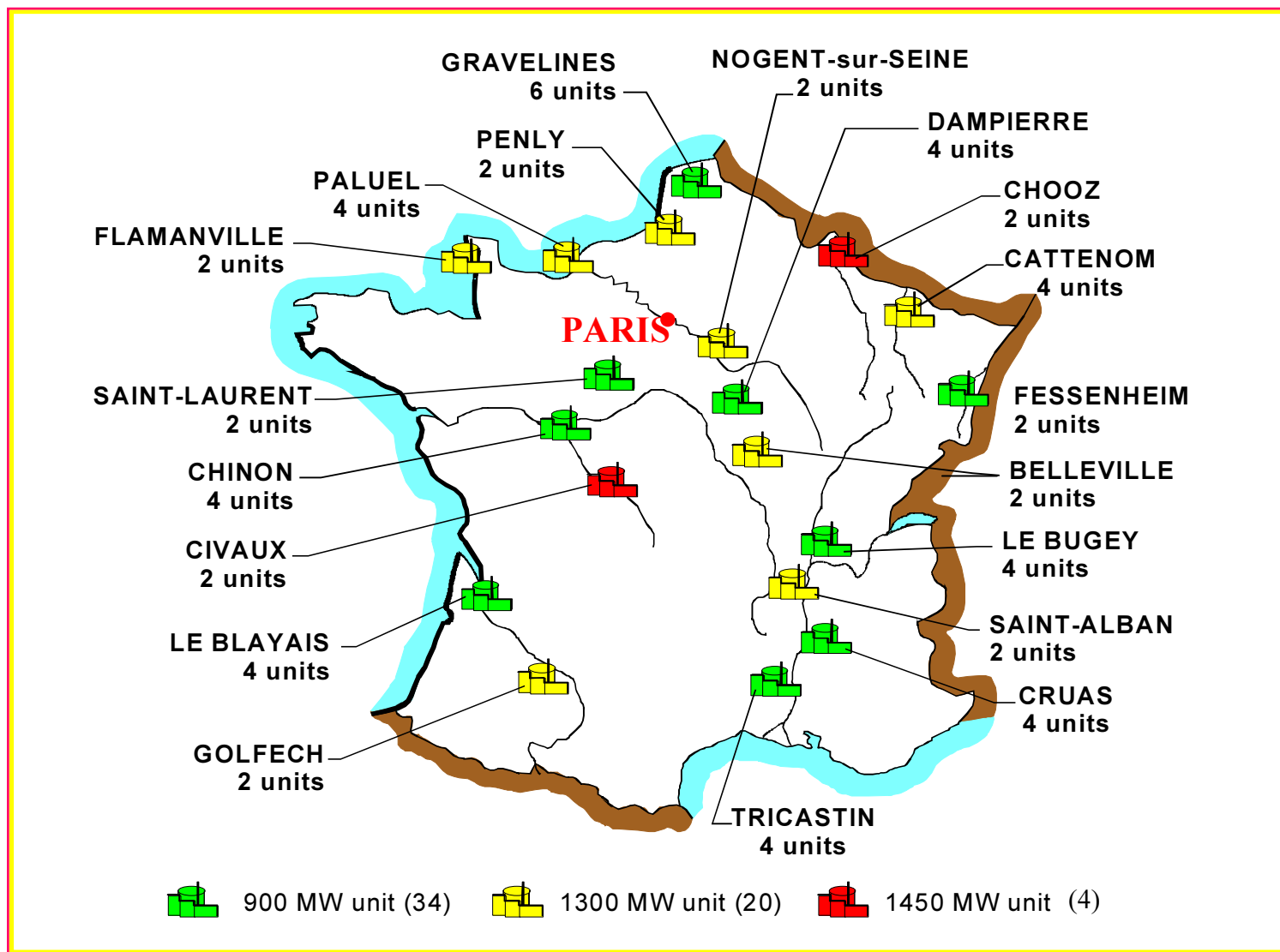
The Nuclear Plants Life Management Policy will take into account the category of the components and structure ; and strategic choices in relation with prediction of Aging Management



REACTOR PRESSURE VESSEL LIFE MANAGEMENT AT FRENCH PWR PLANTS

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PWR in FRANCE.





REACTOR PRESSURE VESSEL INTEGRITY ASSESSMENT

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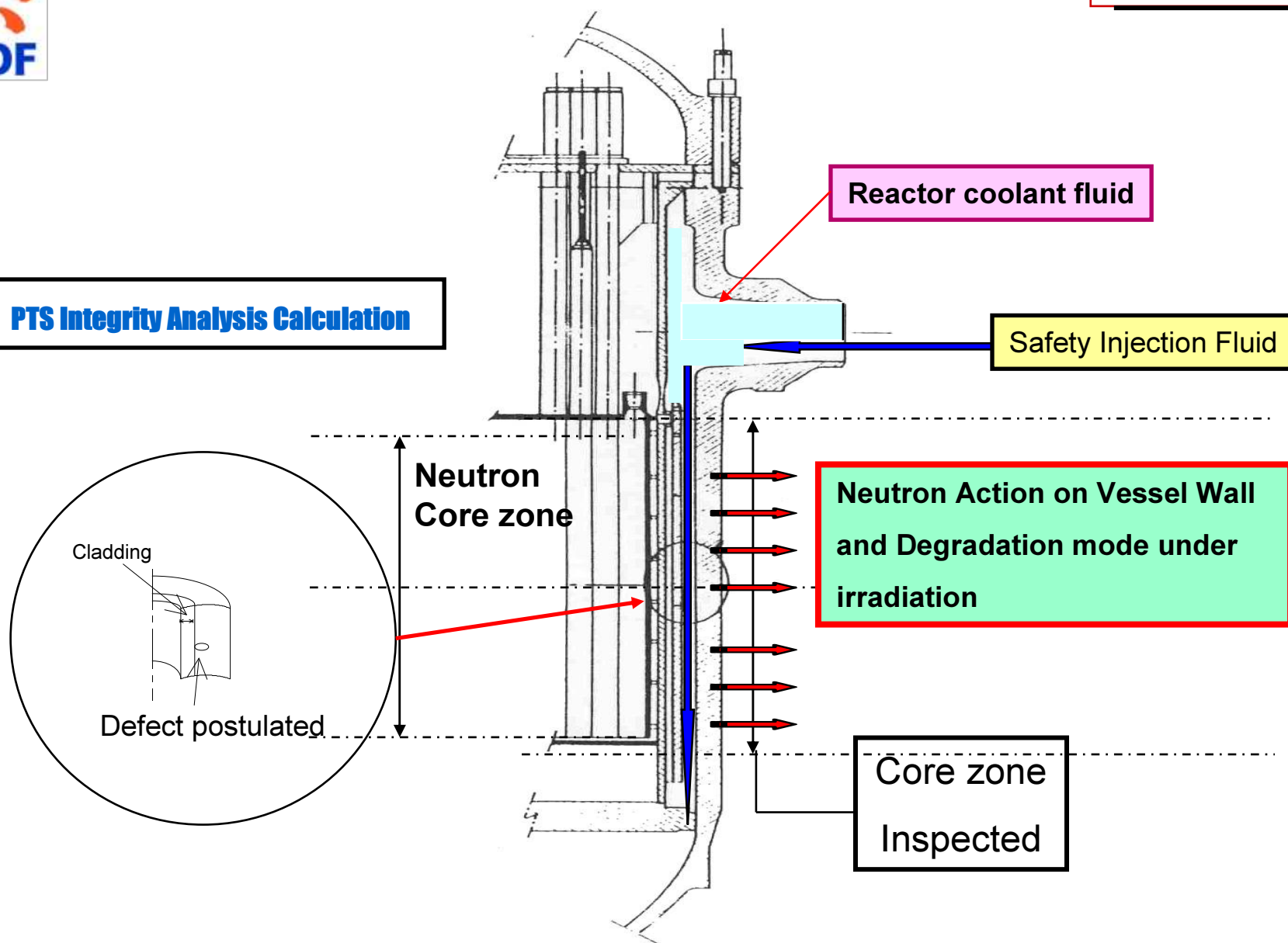
The lifetime of a nuclear power plant may be affected by three main factors :

- ❖ normal wear of its components and systems - called aging - that depends particularly on their age, operating conditions and maintenance conditions applied to them ;
- ❖ the safety level, that must consistently comply with the safety requirements applicable to the power stations at all times, and which could change as a function of new regulations ;
- ❖ competitiveness, which must remain satisfactory compared with the competitiveness of other production means.

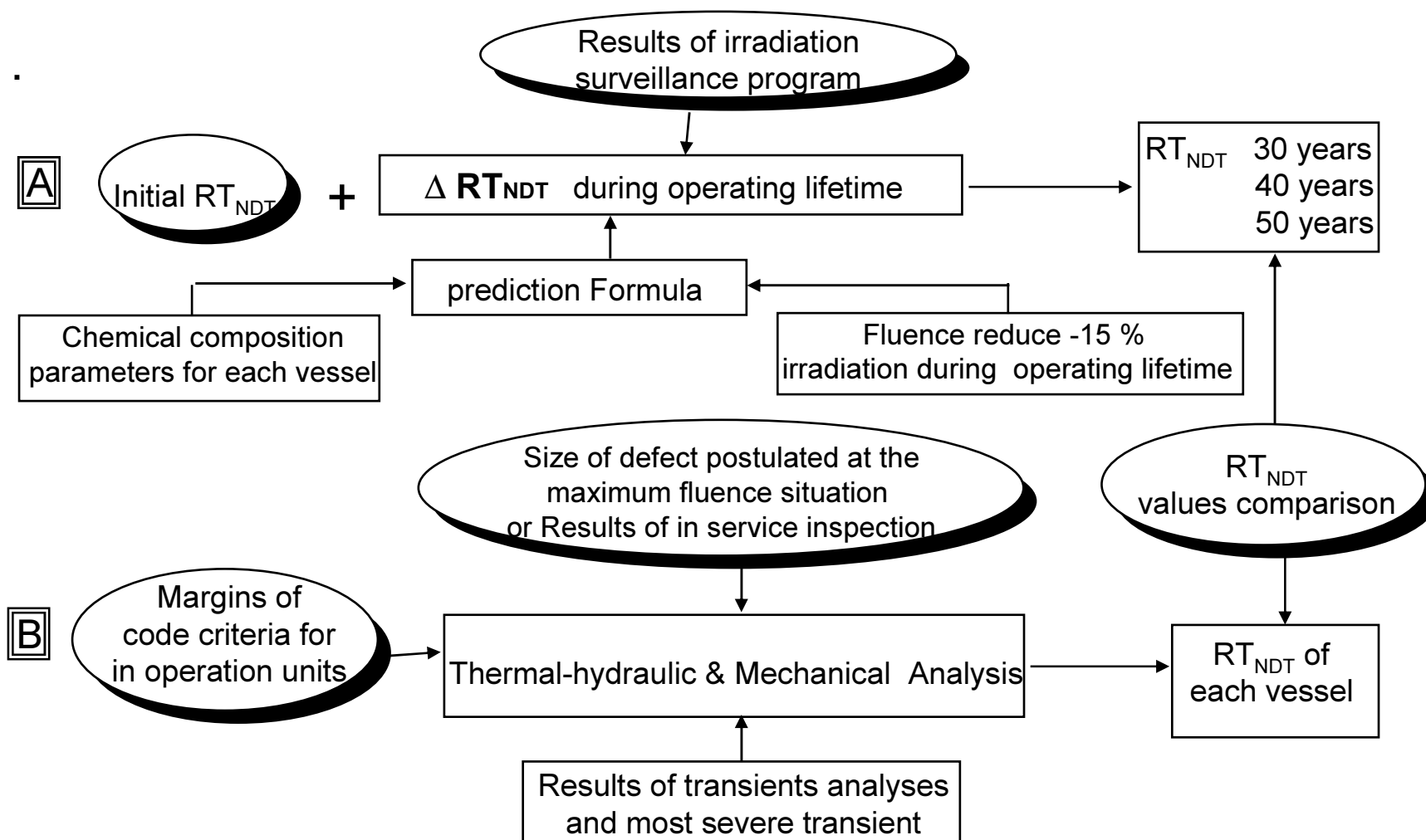
The general strategy is based on the following two points :

- ❖ the ten-year safety reassessment process for each ten years period PSR,
- ❖ the implementation of two structured programmes in order to make sure that all technical and industrial actions necessary to achieve a lifetime of at least 40 years are actually implemented .

PTS Integrity Analysis Calculation

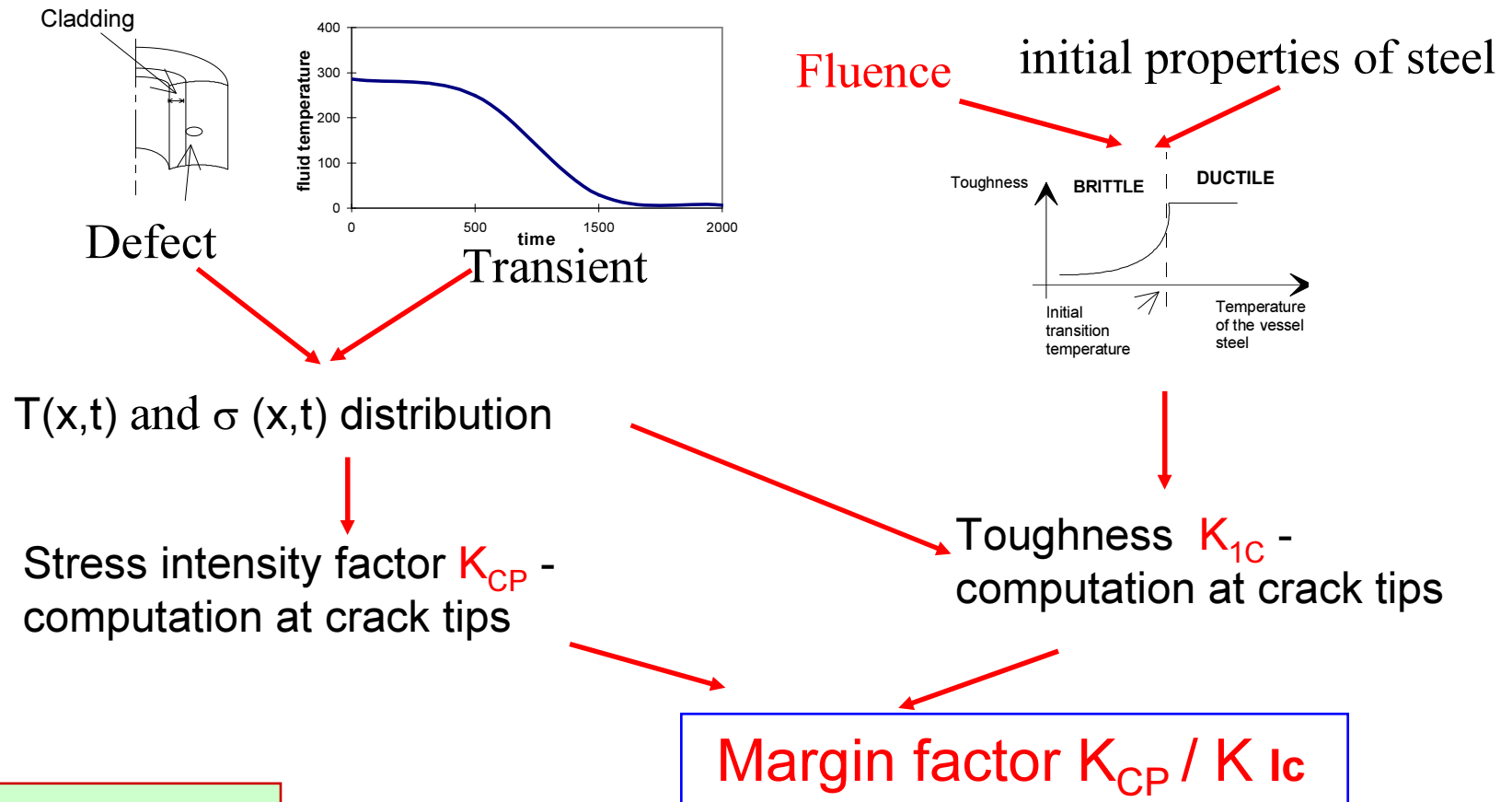


EVALUATION APPROACHES CARRIED OUT SPECIFIC VESSEL by VESSEL EVALUATION



Assessment of RPV integrity : fluence calculations at EDF

Level A = 2nd category of situation $\Rightarrow Sc = 2$
level C = 3rd category of situation $\Rightarrow Sc = 1.6$
Level D = 4th category of situation $\Rightarrow Sc = 1.2$



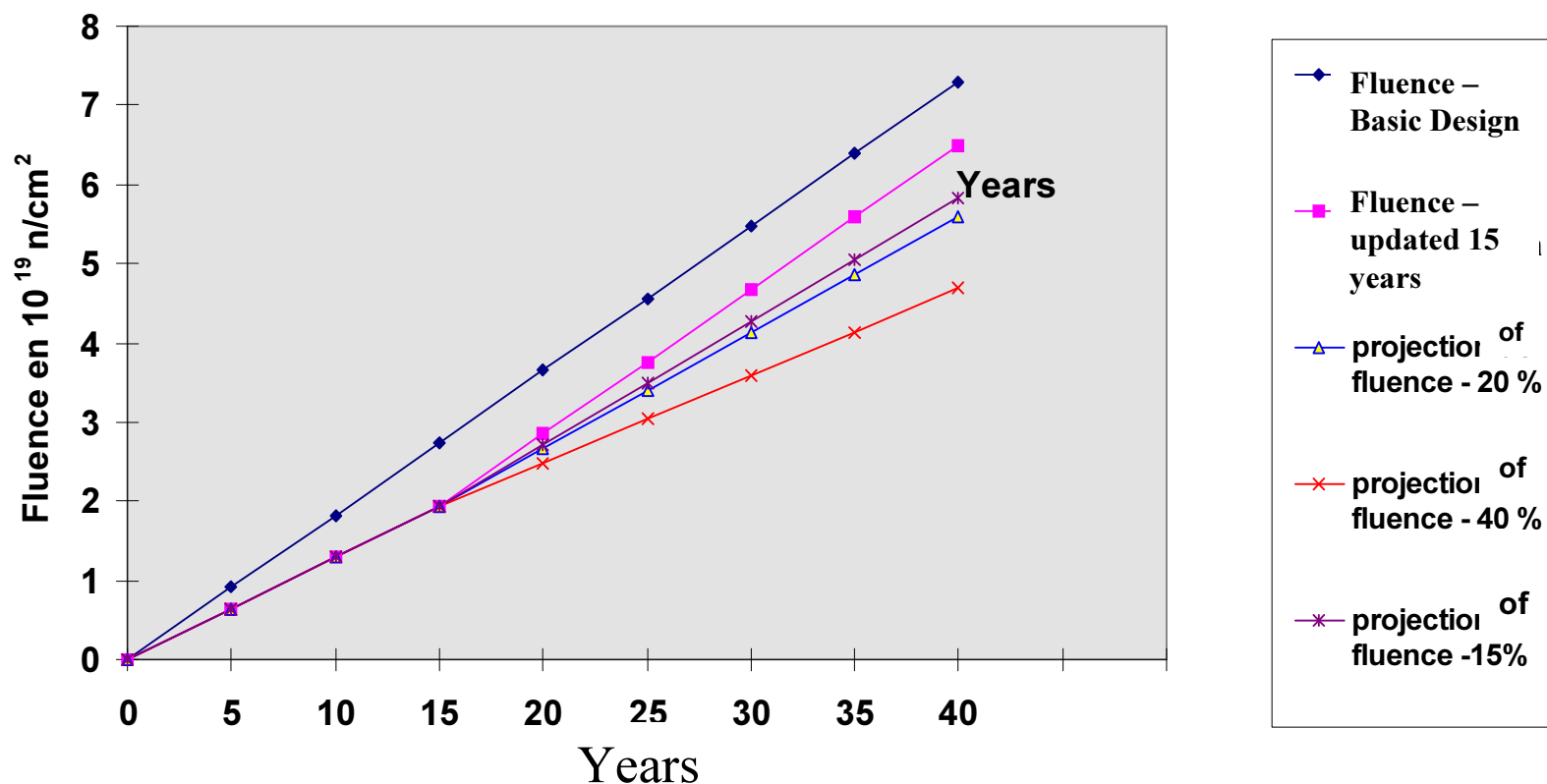
$K_{CP} = K_I$ plastic correction



2 - FLUENCE

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FIGURE 3 : PROJECTIONS DE FLUENCE A 40 ANS



The actual fluence for 40 years lifetime (considering the reduction of fluence) is :

- 3 loop Reactors : **$6.5 \times 10^{19} \text{ n/cm}^2$**
- 4 - Loop RPV (1300 MWe) fluence at 40 years lifetime : **$4.65 \times 10^{19} \text{ n/cm}^2$**



RT_{NDT} VALUES - RPV SURVEILLANCE PROGRAMME

Capsule V

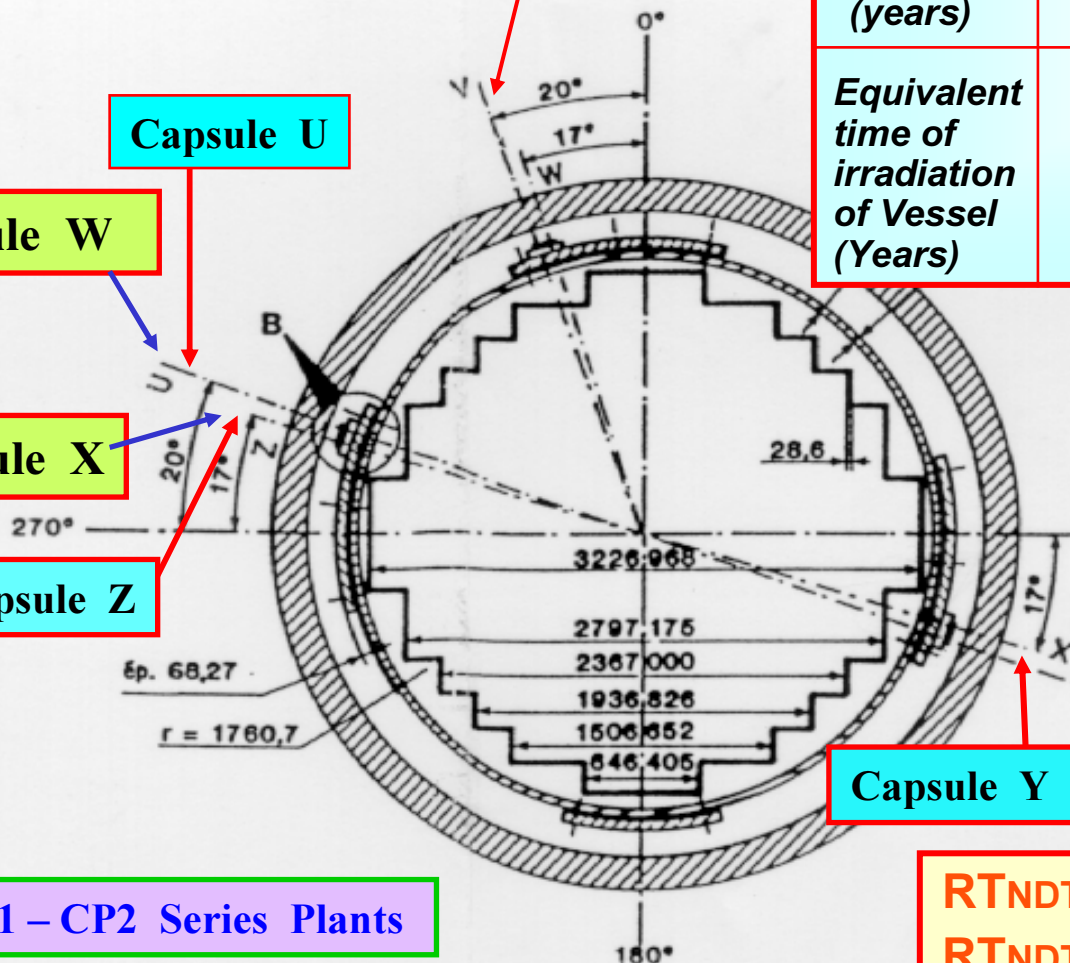
Capsule U

Capsule W

Capsule X

Capsule Z

CP1 – CP2 Series Plants



	Capsule 1 (<i>u</i>)	Capsule 2 (<i>v</i>)	Capsule 3 (<i>z</i>)	Capsule 4 (<i>y</i>)
<i>Time of Duration in vessel (years)</i>	4	7	9	14
<i>Equivalent time of irradiation of Vessel (Years)</i>	11.2	19.5	28.0	40

Extension of the RPV irradiation surveillance : introduction of reserve irradiation capsules in the French plants since 1999 for all reactors.

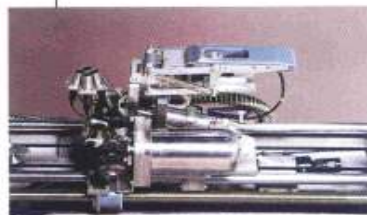
Two reserve capsules **W** and **X** in place of capsules **U** and **Z** after removing from reactor these capsules **U** and **Z**.

Capsule Y

RT_{NDT} = 82 °C -> CP0 units
RT_{NDT} = 73°C -> CP1/CP 2 units.



Outil MIT



Outil RDC (zones réparées des tubulures)



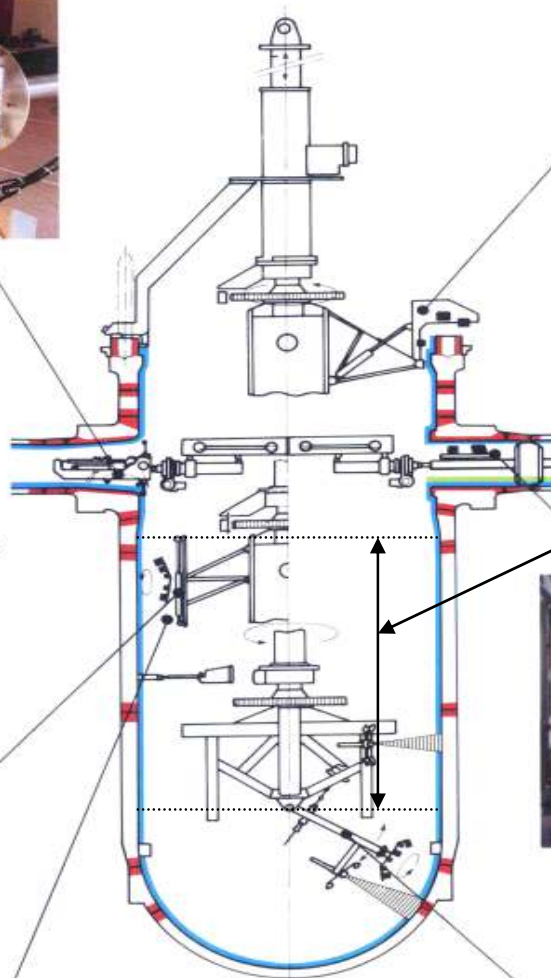
Outil VIROLE



Outil TPM



Outil BRIDE



CORE ZONE INSPECTION



Outil SUPER GAMMA (RT + UT)

Outil FOND SPHERIQUE



VPM



STEAM GENERATOR REPLACEMENT PROGRAMME and STRATEGY

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The global **Life Management strategy** and **Maintenance Program** of **Steam Generators** are in relation with:

- The importance of the inspection Programme
- The on-going actions during NPPs time in operation normal and current Maintenance
- To anticipate the exceptional maintenance
- To anticipate to order the strategic components

Classification of plants with SG Replacement strategy

PERIODIC INSPECTIONS OF STEAM GENERATOR

- Technical SG End of Life – SG **Plugging Rate max: 12,5 or 15%**
- Planning schedule in function of intervention on Plants (industrial capability, Availability of reserve of SG spare parts)
- Connection with SGR and 10-years outages



Classification of PWR plants in relation with SR Replacement strategy

➤ Group 1

Units with SG affected by dégradations and SGR is will be carry out in futur (date of SGR estimated)

➤ Group 2

Units with SG lifetime evaluation is with incertainties (evolution of dégradations is unknown, treat on SGR Planed with 3rd 10-years outage

➤ Group 3

Units with SG Lifetime = Plant lifetime

GROUP 1

**DAMPIERRE 2, DAMPIERRE 4
GRAVELINES 3
BUGEY 2, BUGEY 3, BUGEY 4
FESSENHEIM 2
BLAYAIS 1**

GROUP 2

**CHINON B1, CHINON B2
BLAYAIS 2, BLAYAIS 3,
BLAYAIS 4**

GROUP 3 900 Mwe : 21 units

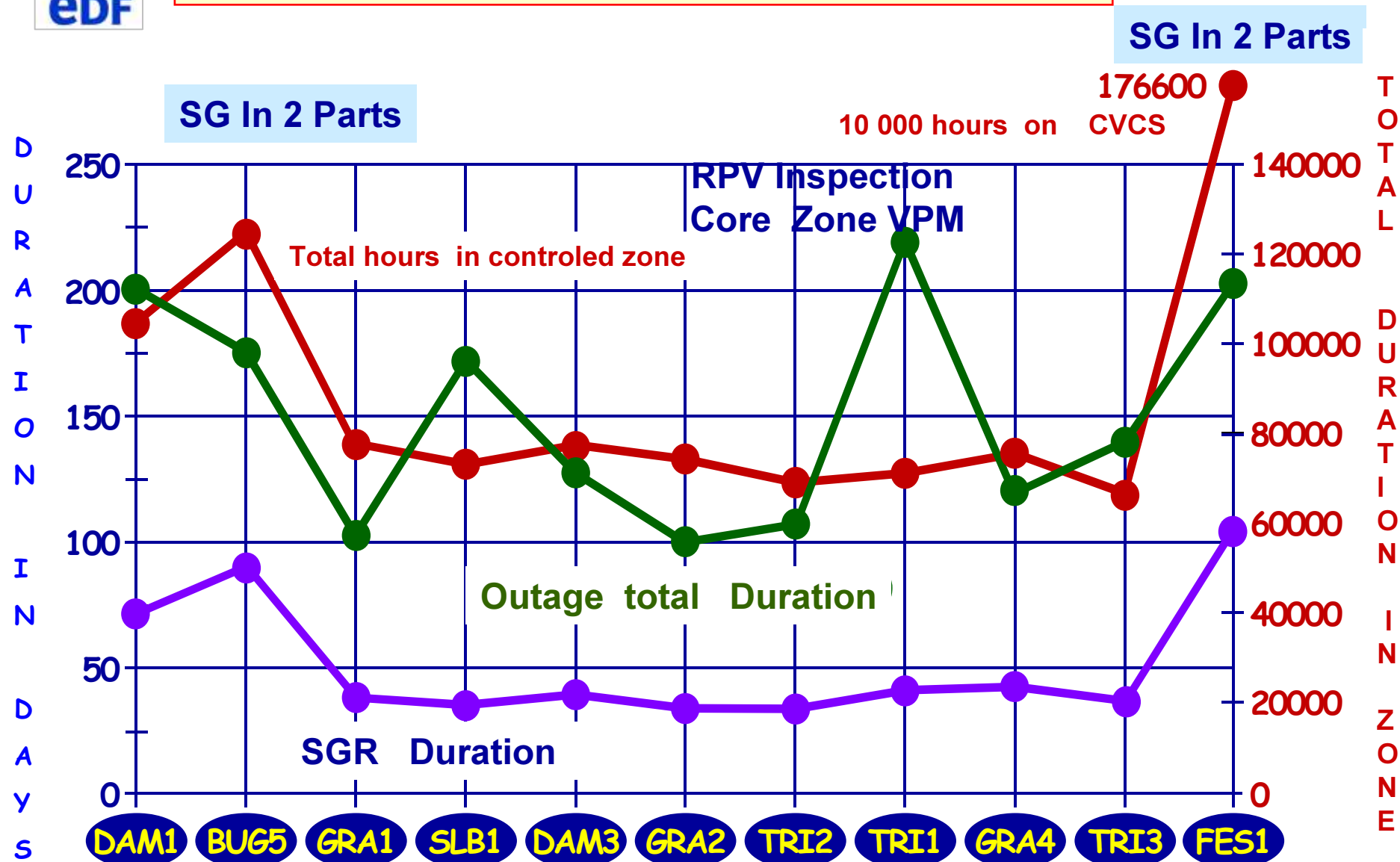
**CRUAS 1-2-3-4
GRAVELINES 5 – 6
CHINON B3 – B4**

- Units with SG replaced in past
- 1300 MWe: all 20 units
- 1450 MWe : all 4 units



SYNTHESIS of DURATION FRENCH SGR

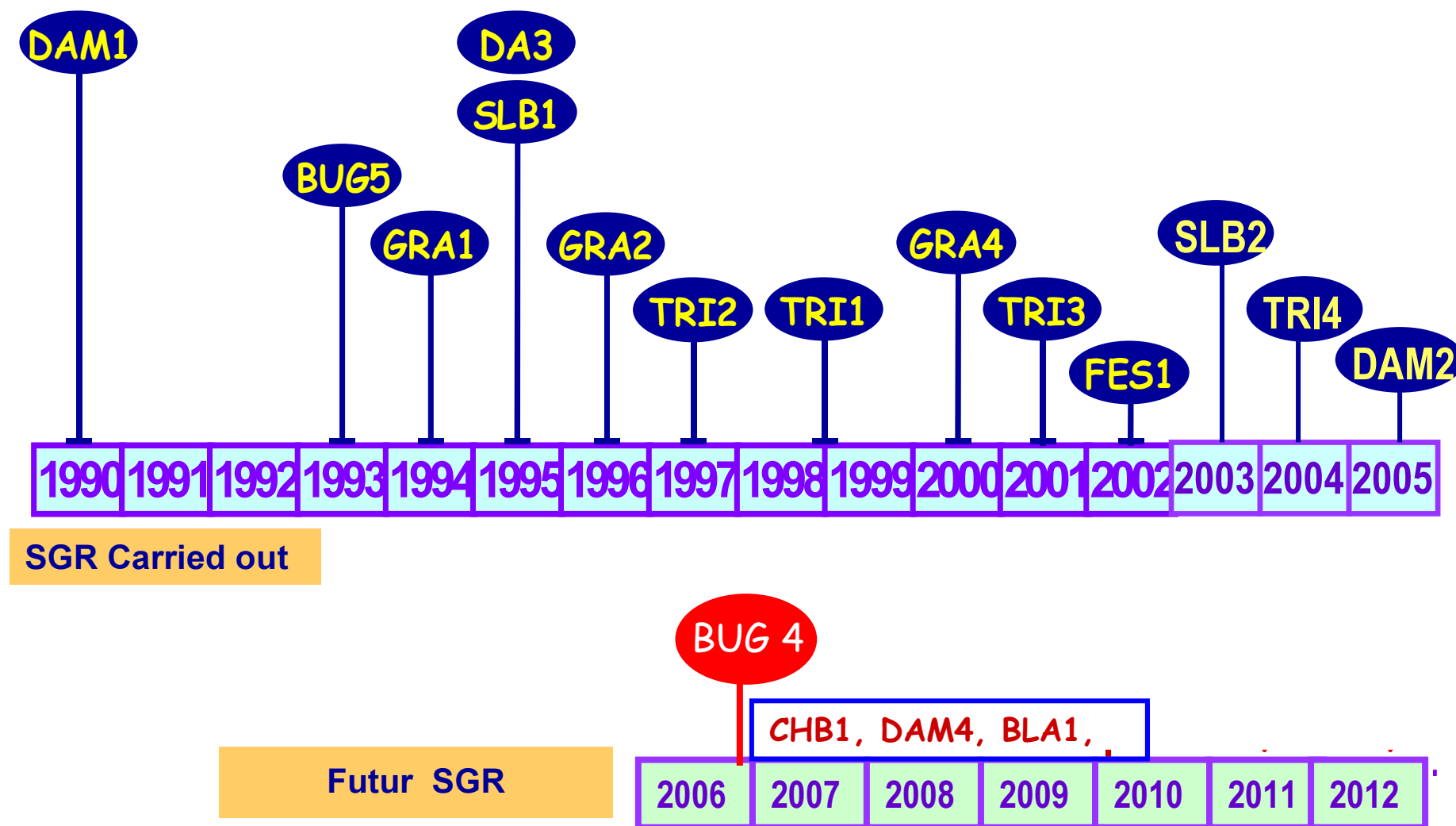
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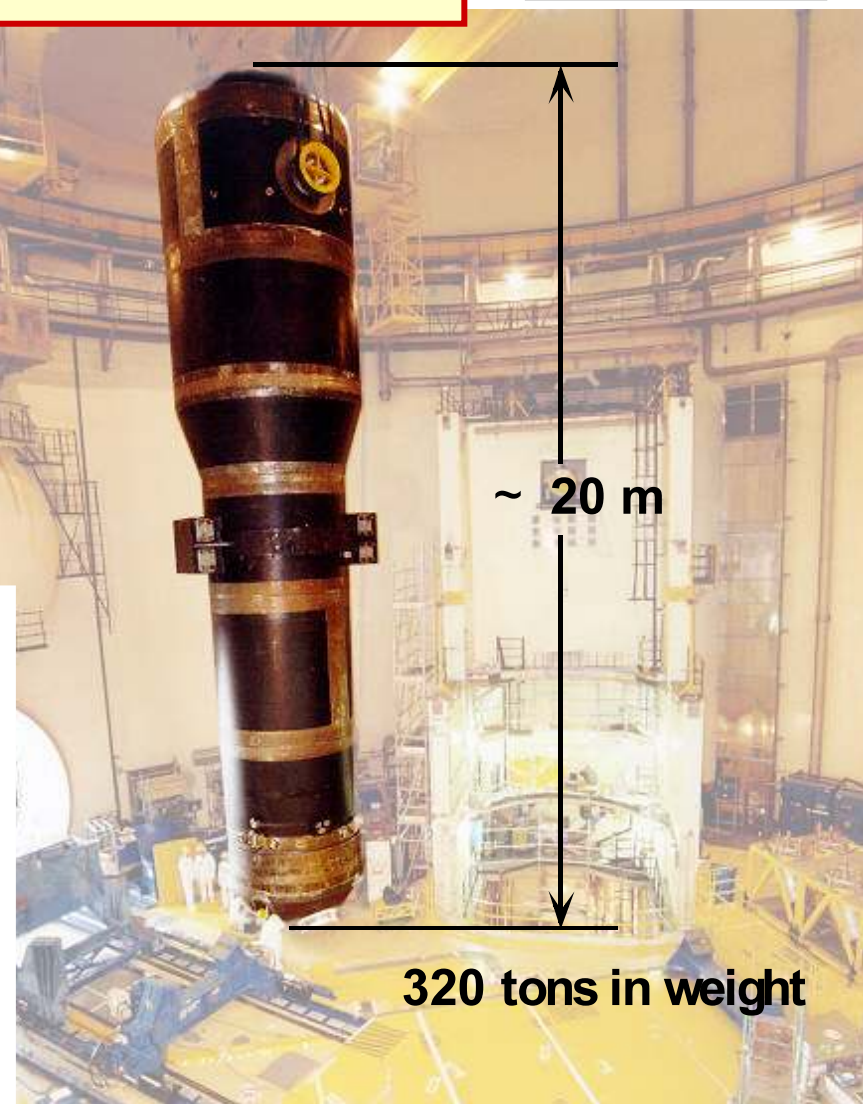


PLANNING OF SG REPLACEMENT

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Example SGR One Piece Monobloc



CONCLUSION

All of actions engaged and applied on 58 NPPs taking into account the large informations capitalised in database from the feedback experience informations , current Maintenance and exceptionnal maintenance give to the french utility the good approaches to the long term life Management

The importance of Criteria combined with periodic inspection and in relation with:

- The global Maintenance program and strategy
Specific Maintenance and Inspection for **RPV** and for **Steam Generator**
- The monitoring to enrich the feedback experience and database collected on Steam Generator and RPV
- To anticipate the exceptional maintenance
Mainly for SGR
- To anticipate to order the strategic components
are very important for Plant Life Mnagement Strategy